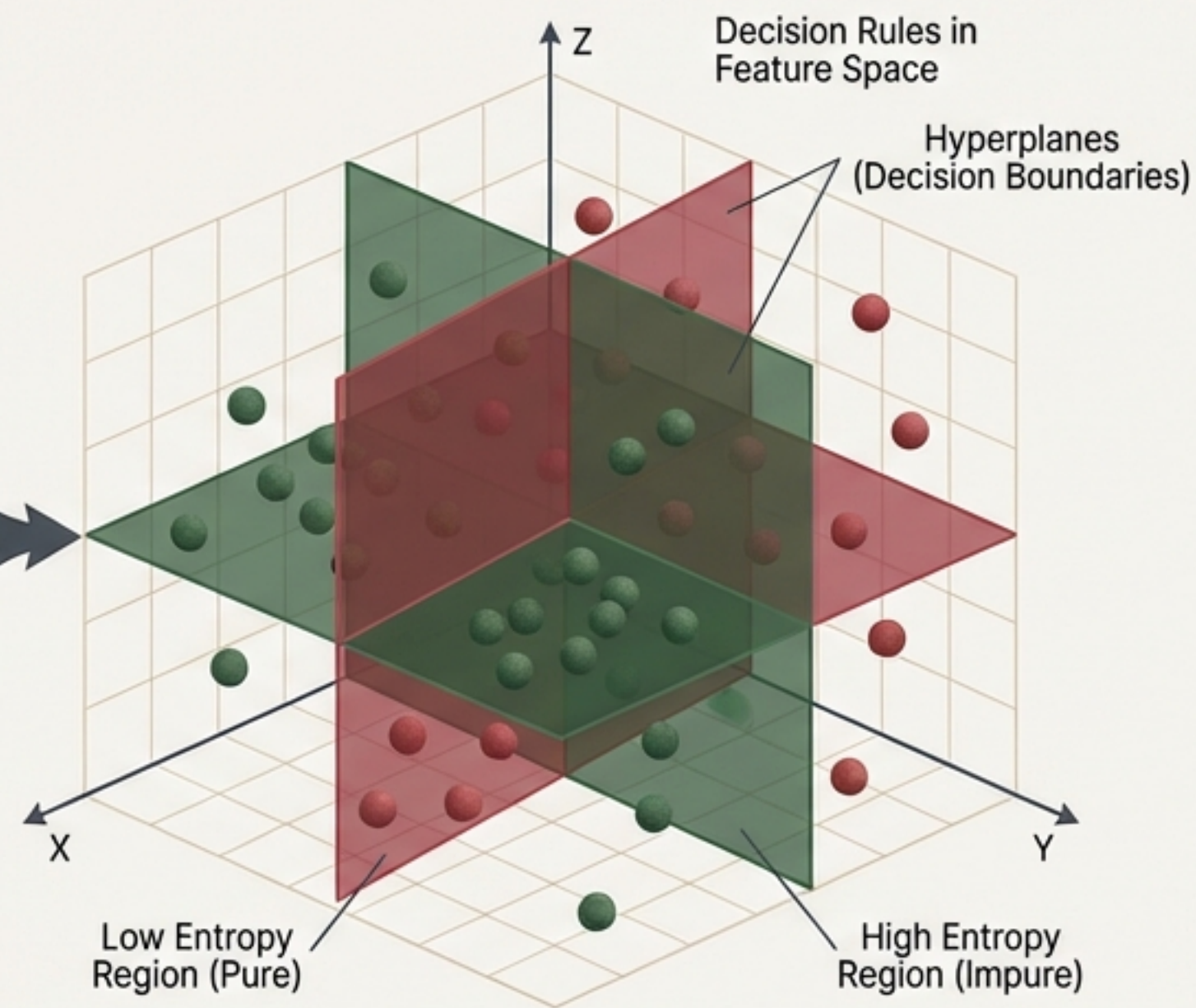
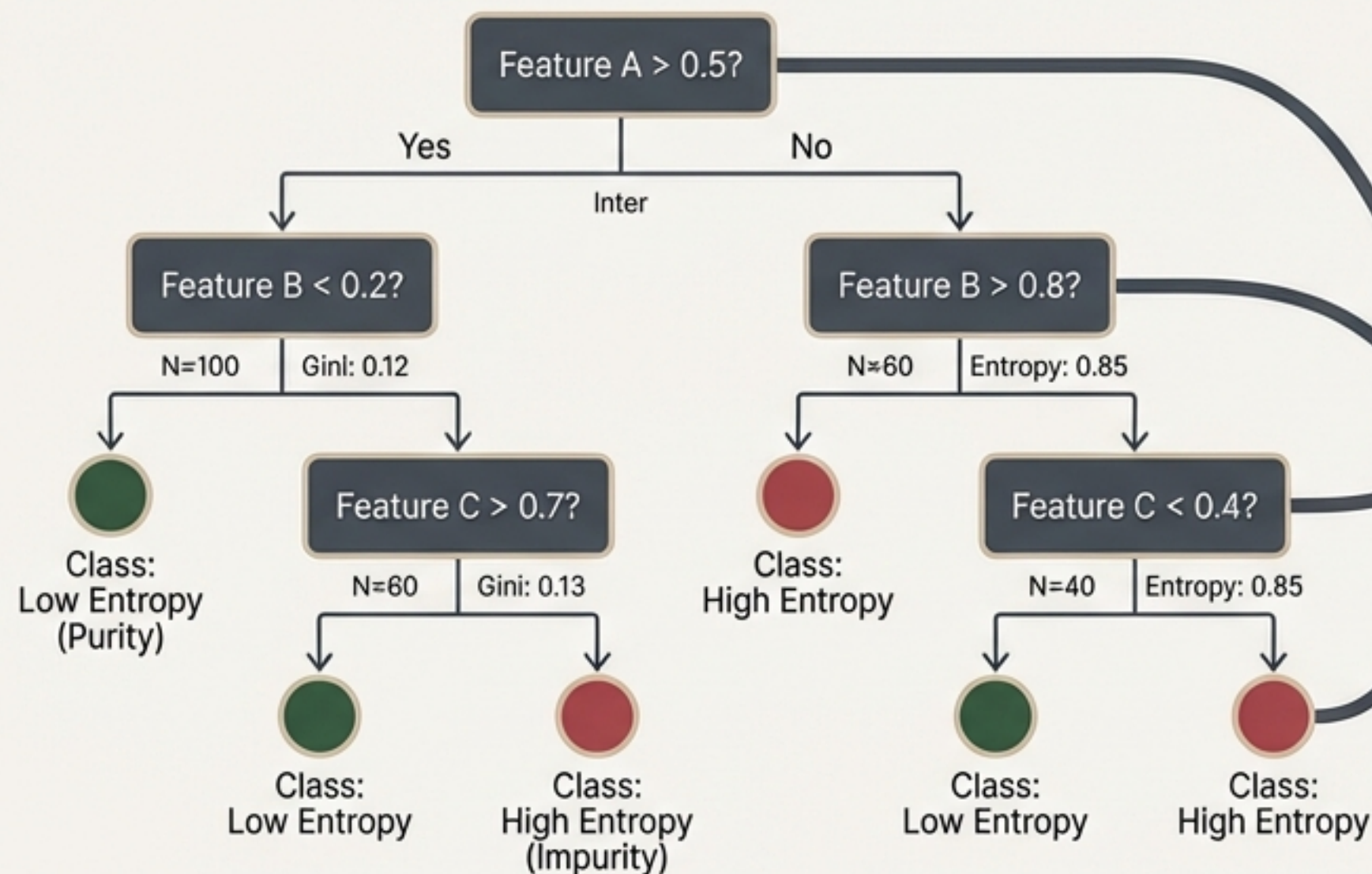


Demystifying Decision Trees

From Geometric Intuition to Mathematical Purity.

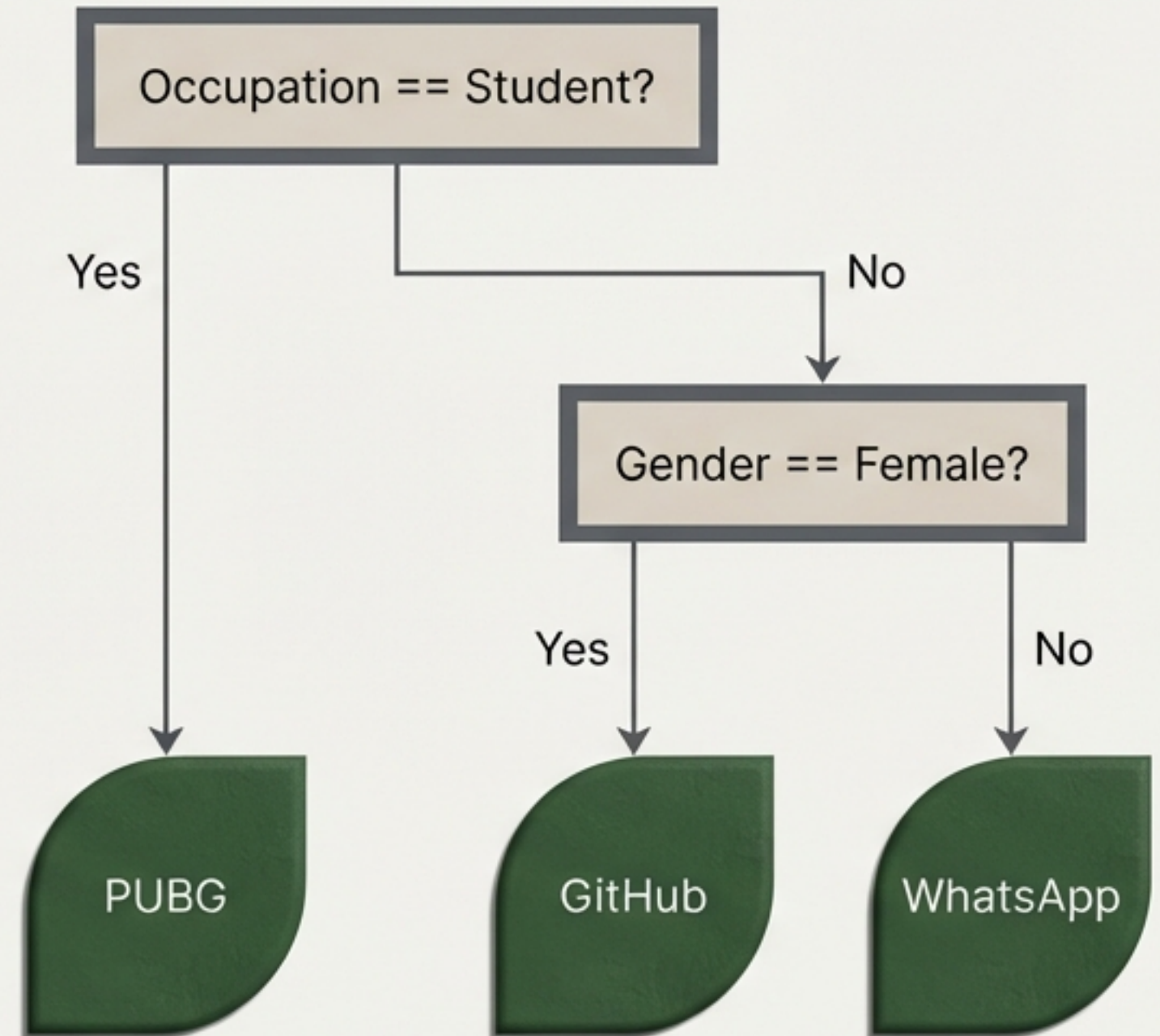


Decision Trees: Hierarchical structure of sequential rules splitting data into progressively purer subsets based on feature values.

Geometric Interpretation: Each split in the tree corresponds to a hyperplane dividing the feature space into hypercuboids containing homogeneous data.

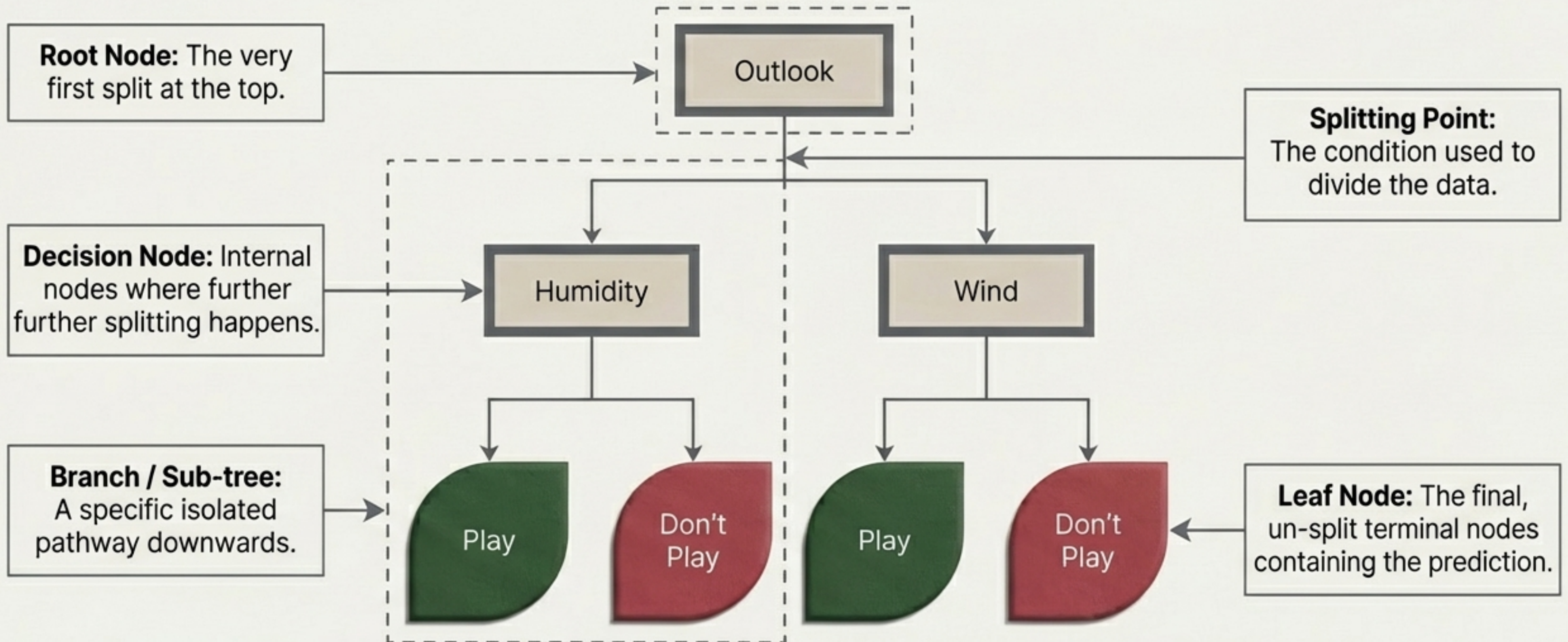
The '20 Questions' Framework

Decision Trees are essentially automated, nested If-Else statements. They recursively ask the best possible questions to split data into pure categories.



The Anatomy of a Tree

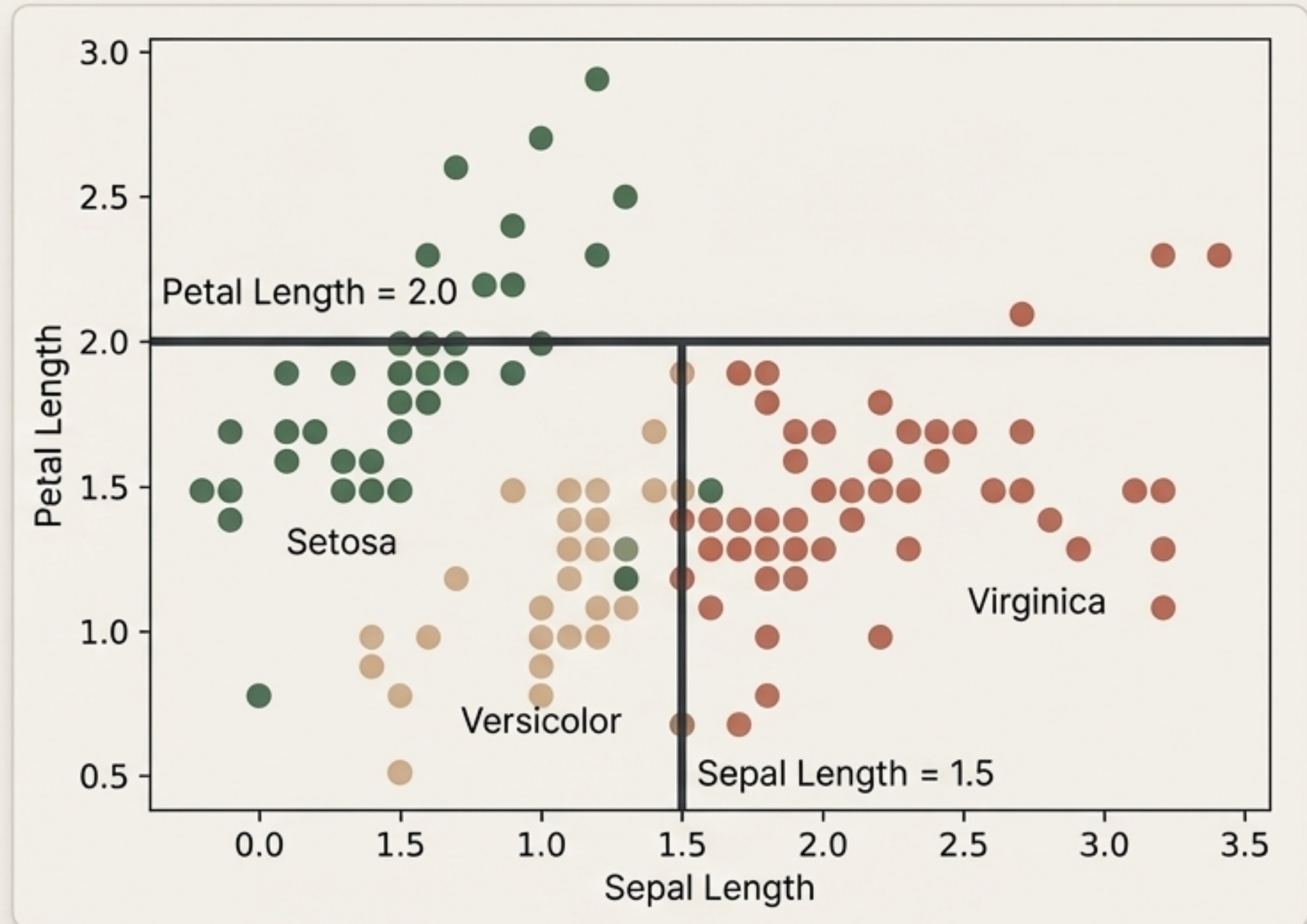
Decision trees operate by recursively asking questions to split data, creating a flowchart-like structure where each path leads to a prediction. This structure is composed of specific elements that define its operation.



The Geometric Reality: Drawing Hyperplanes

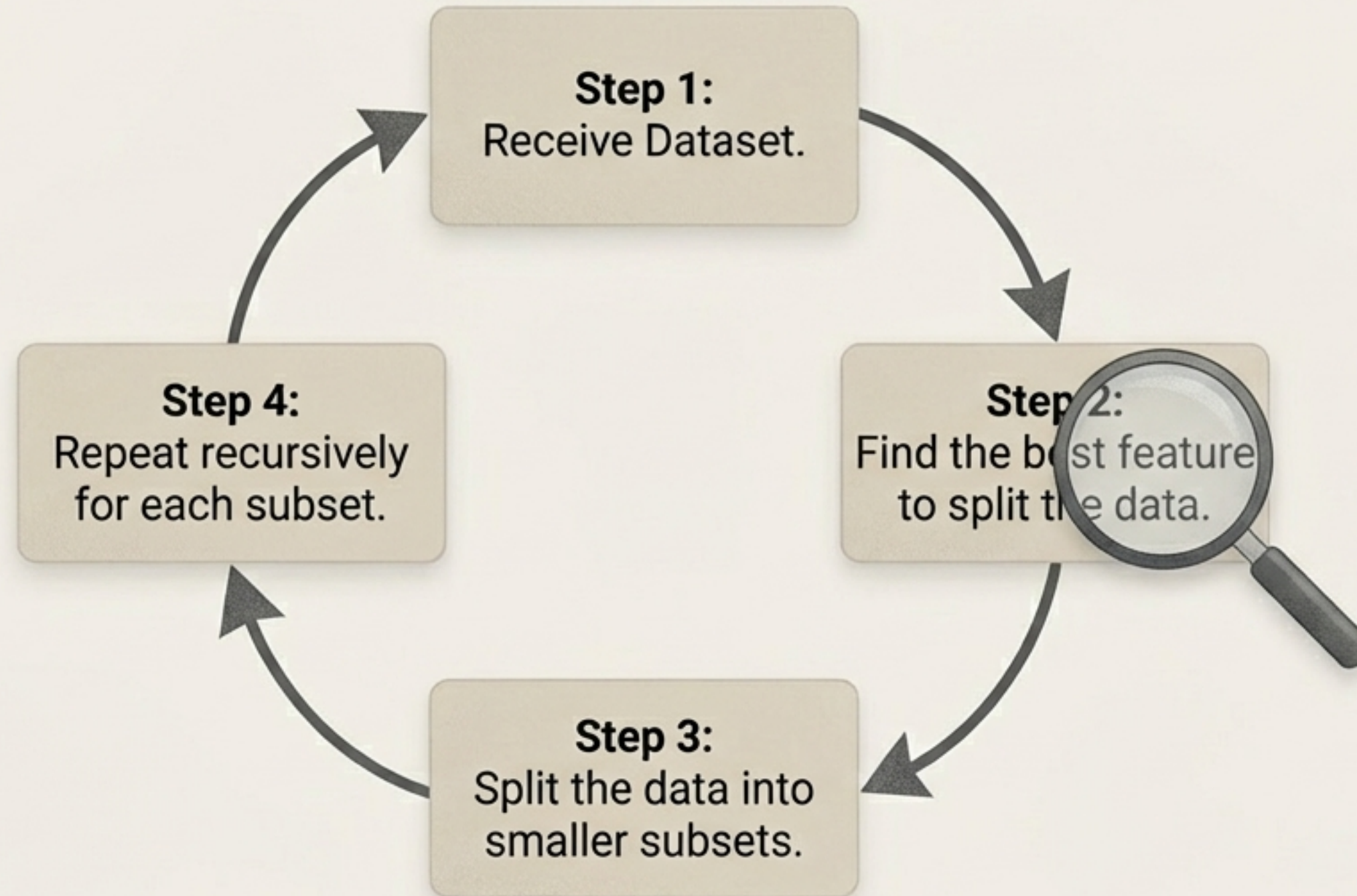
Mathematically, a Decision Tree draws axis-parallel lines (hyperplanes) to cut coordinate systems. In 2D, these are lines.

In 3D, they are planes slicing through a room. In higher dimensions, they create multi-dimensional hypercuboids.



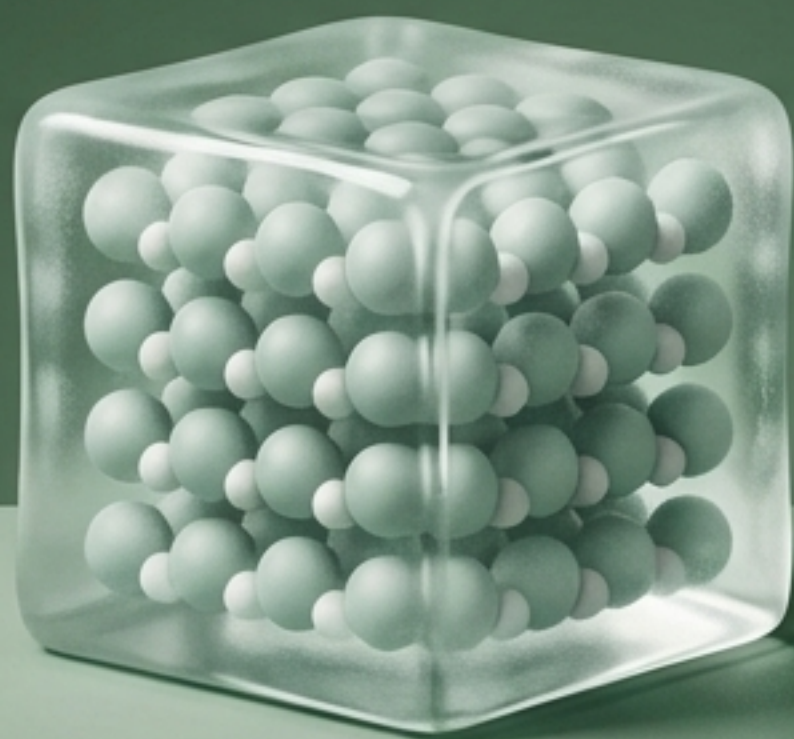
The Recursive Splitting Loop

How do we mathematically define the best feature to split on? We need a metric to measure impurity.



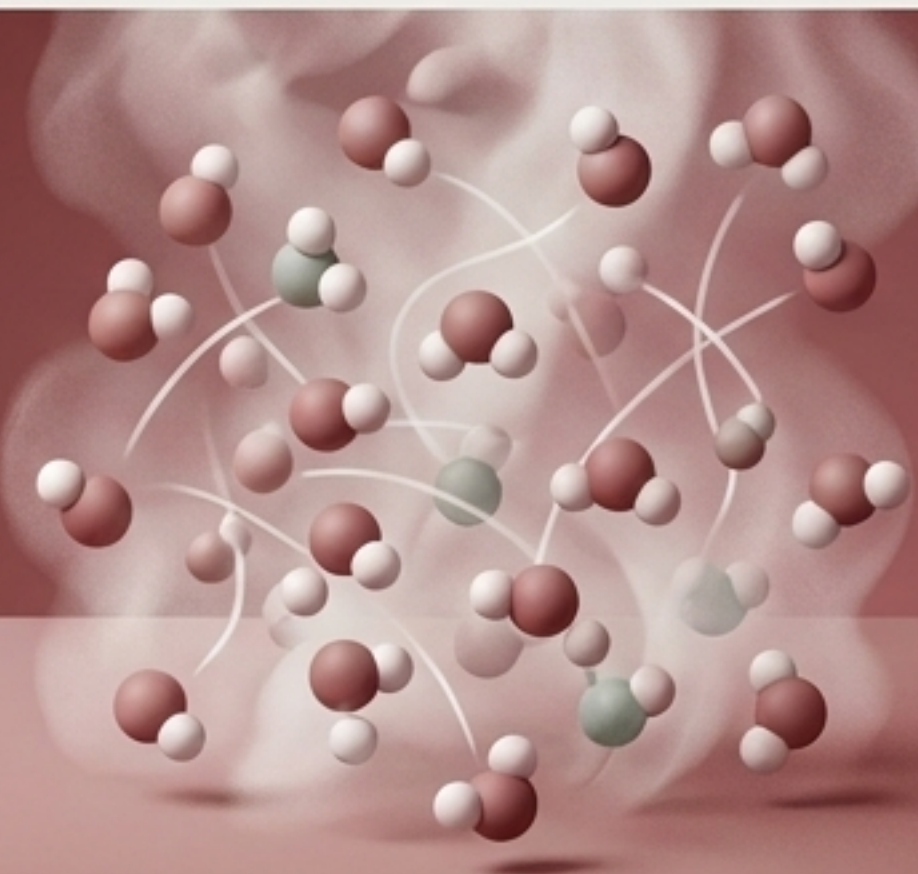
Measuring Purity with Entropy

Entropy measures the lack of order or predictability.
More knowledge about the system equals lower Entropy.



Low Entropy = High Purity

Representing a dataset with 100% of one class.



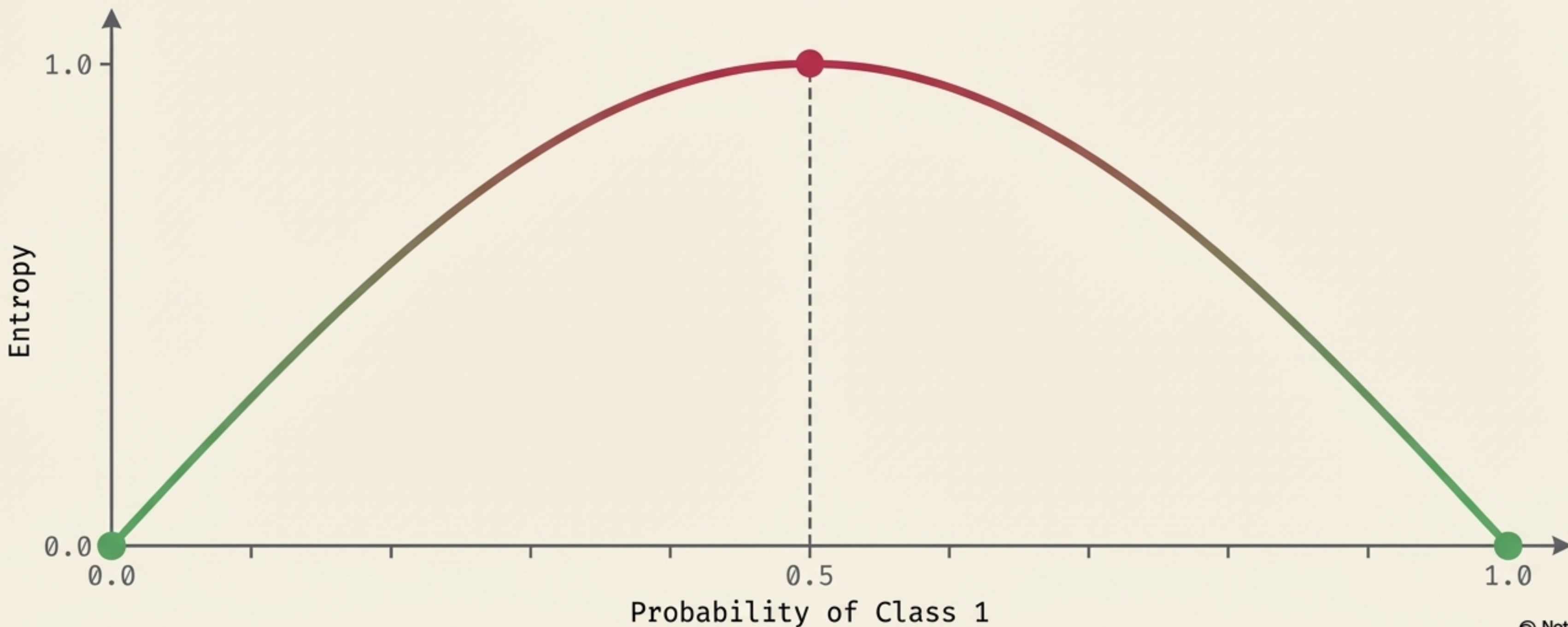
High Entropy = High Uncertainty

Representing a dataset mixed exactly 50/50 between two classes.

The Mathematics of Uncertainty

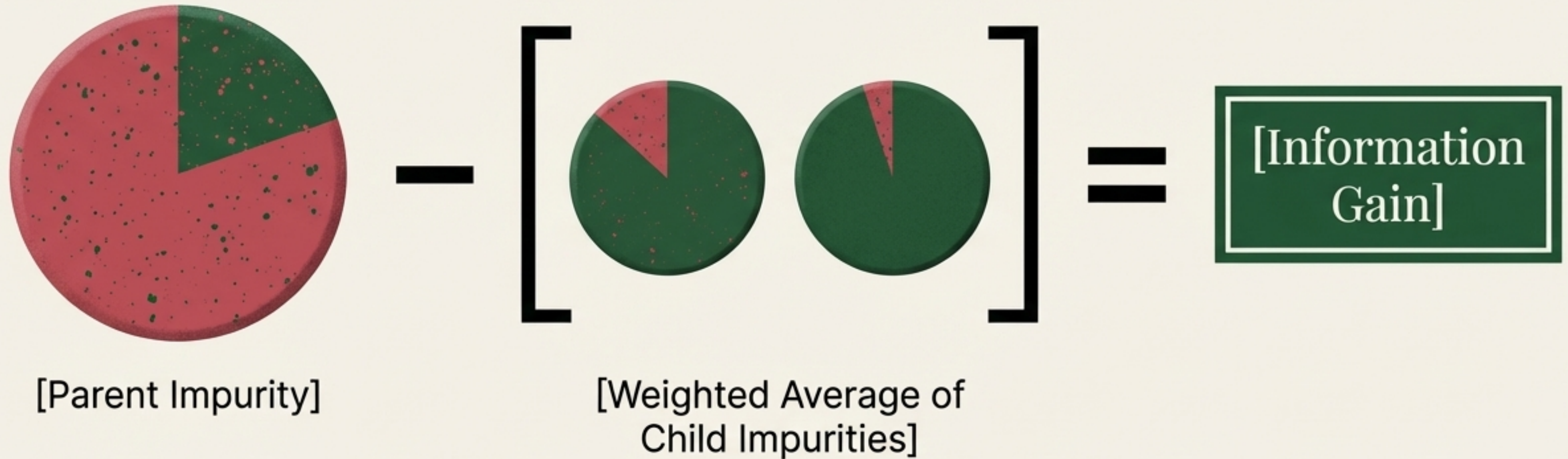
A 50/50 split yields maximum uncertainty (1.0). A 100/0 split yields total certainty (0.0).

$$\text{Entropy} = - \sum p_i \log_2(p_i)$$



The Decision Engine: Information Gain

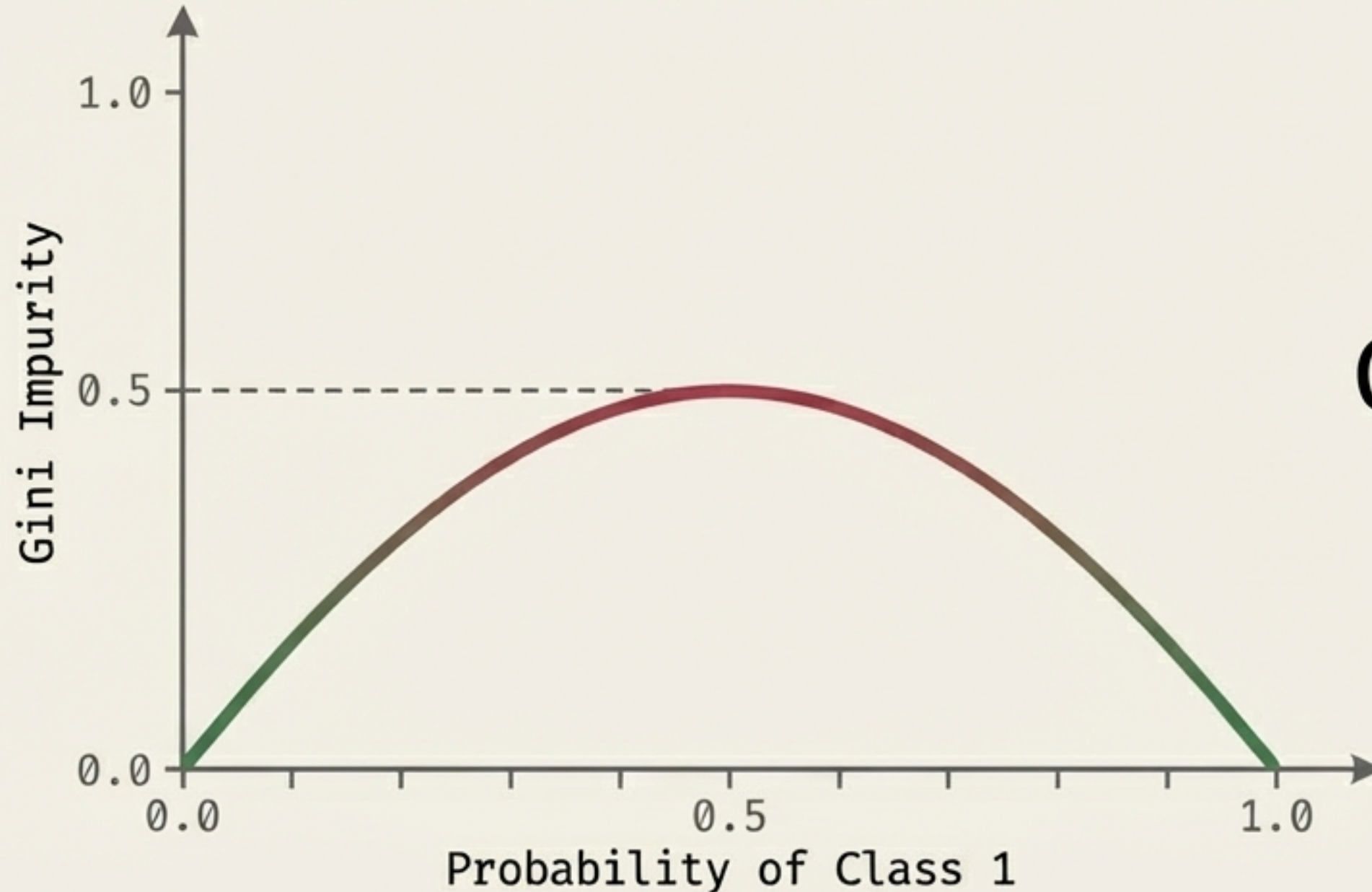
Information Gain evaluates the quality of a split. The algorithm calculates the decrease in Entropy for every possible column and chooses the split that provides the highest Information Gain.



Gini Impurity: The Faster Alternative

Gini Impurity serves the exact same purpose as Entropy but avoids computationally expensive logarithmic calculations by using squares instead.

Minimum value: 0.0 (Pure). Maximum value for binary classification: 0.5 (Maximum Impurity).



$$\text{Gini} = 1 - \sum (p_i)^2$$

Entropy vs. Gini

Both metrics yield near-identical results. Use Gini for computational speed; consider Entropy if your specific dataset requires strictly balanced splits.

Entropy



- Uses Logs ($\log_2$)
- Peaks at 1.0
- Slower, but produces balanced trees

Gini

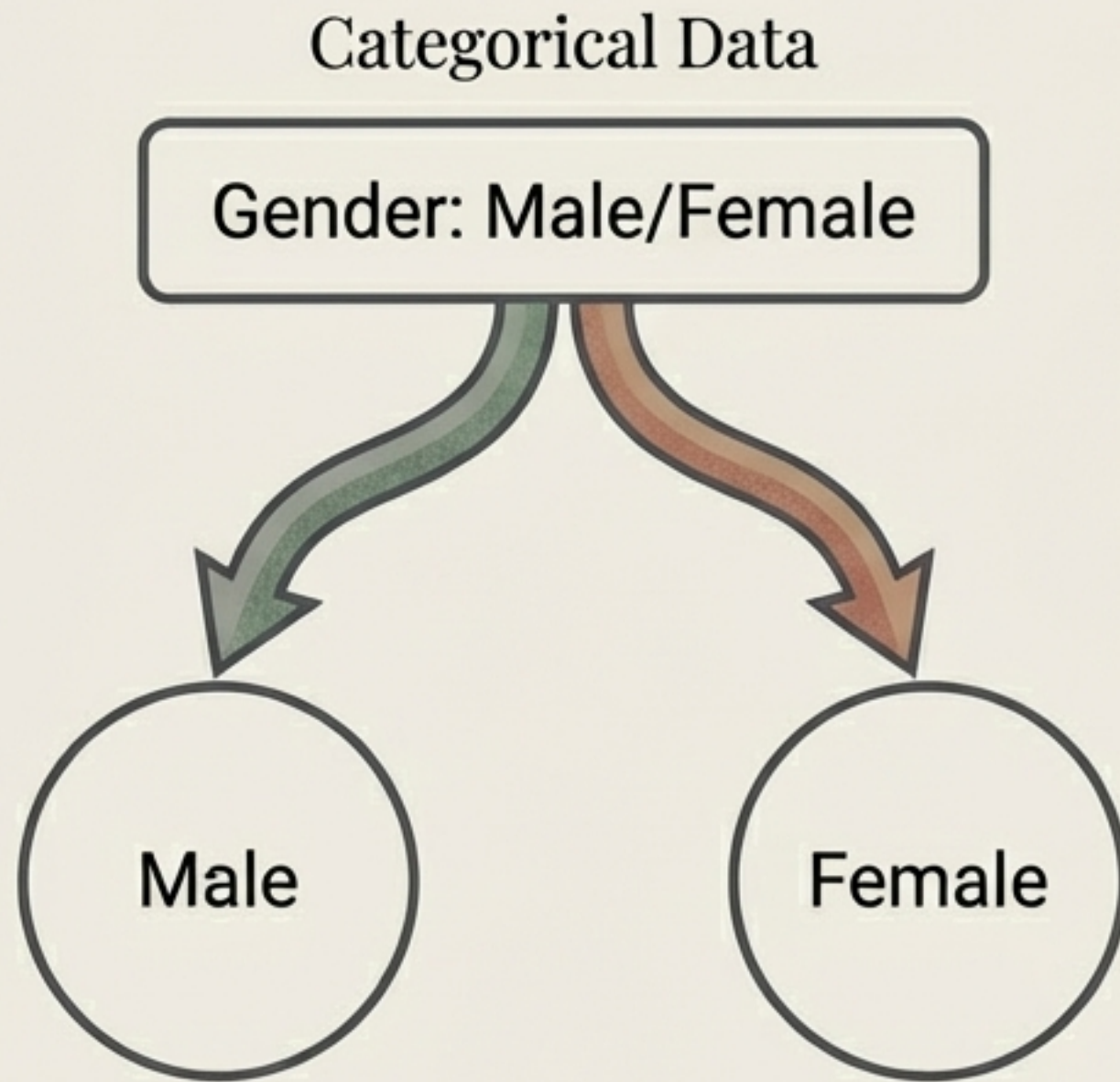


- Uses Squares (p^2)
- Peaks at 0.5
- Faster, default for modern libraries



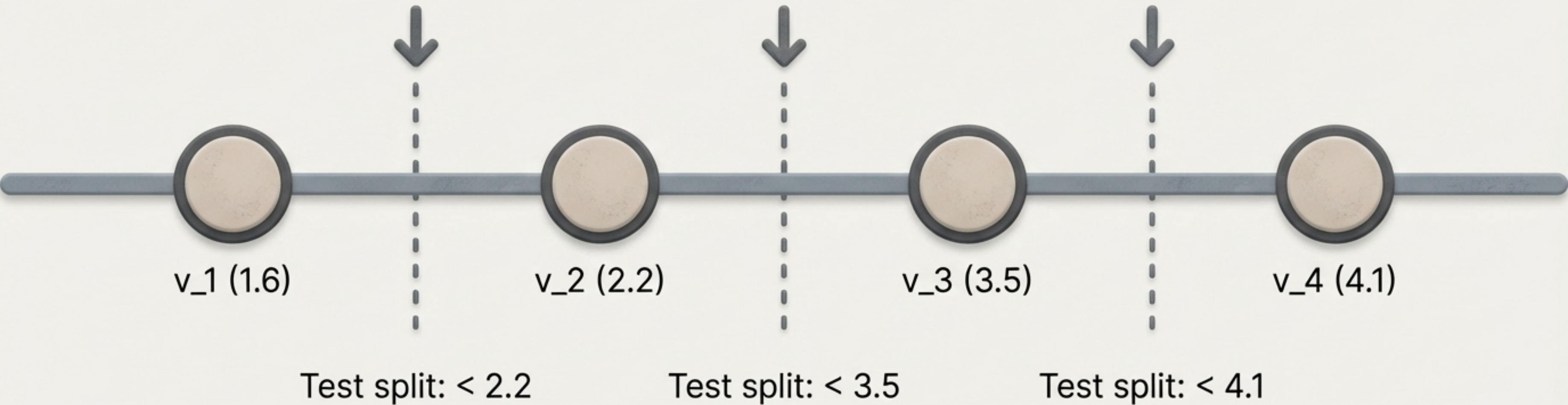
The Numerical Challenge

Categorical variables naturally group data. But how does a tree split a continuous spectrum of numbers without creating a separate branch for every single row?



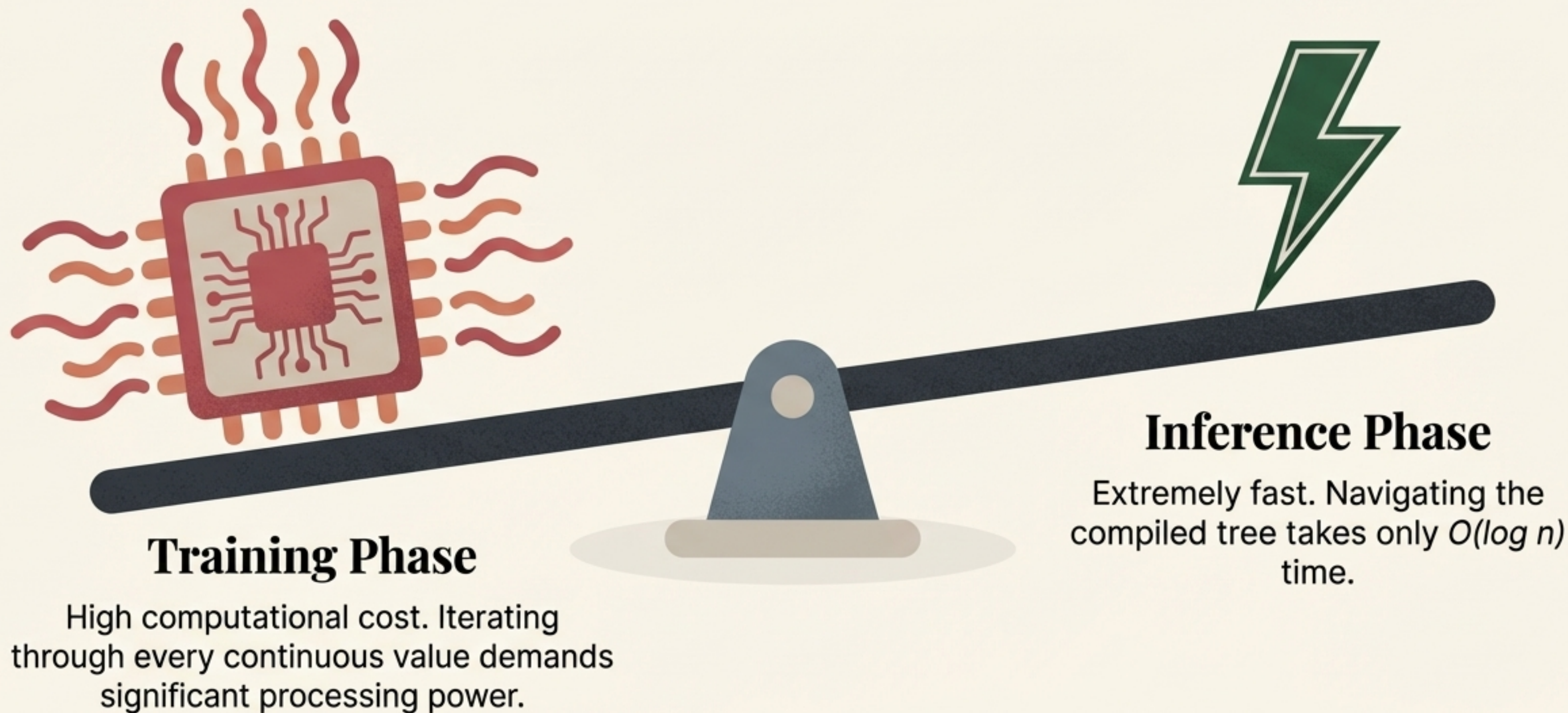
Thresholding Continuous Variables

The algorithm sorts the numerical data and treats every single value as a potential threshold. It calculates the Information Gain for every possible split point and selects the threshold that yields the absolute maximum gain.



Time Complexity Dynamics

Training a tree on numerical data is computationally expensive, but querying the final model is lightning-fast.



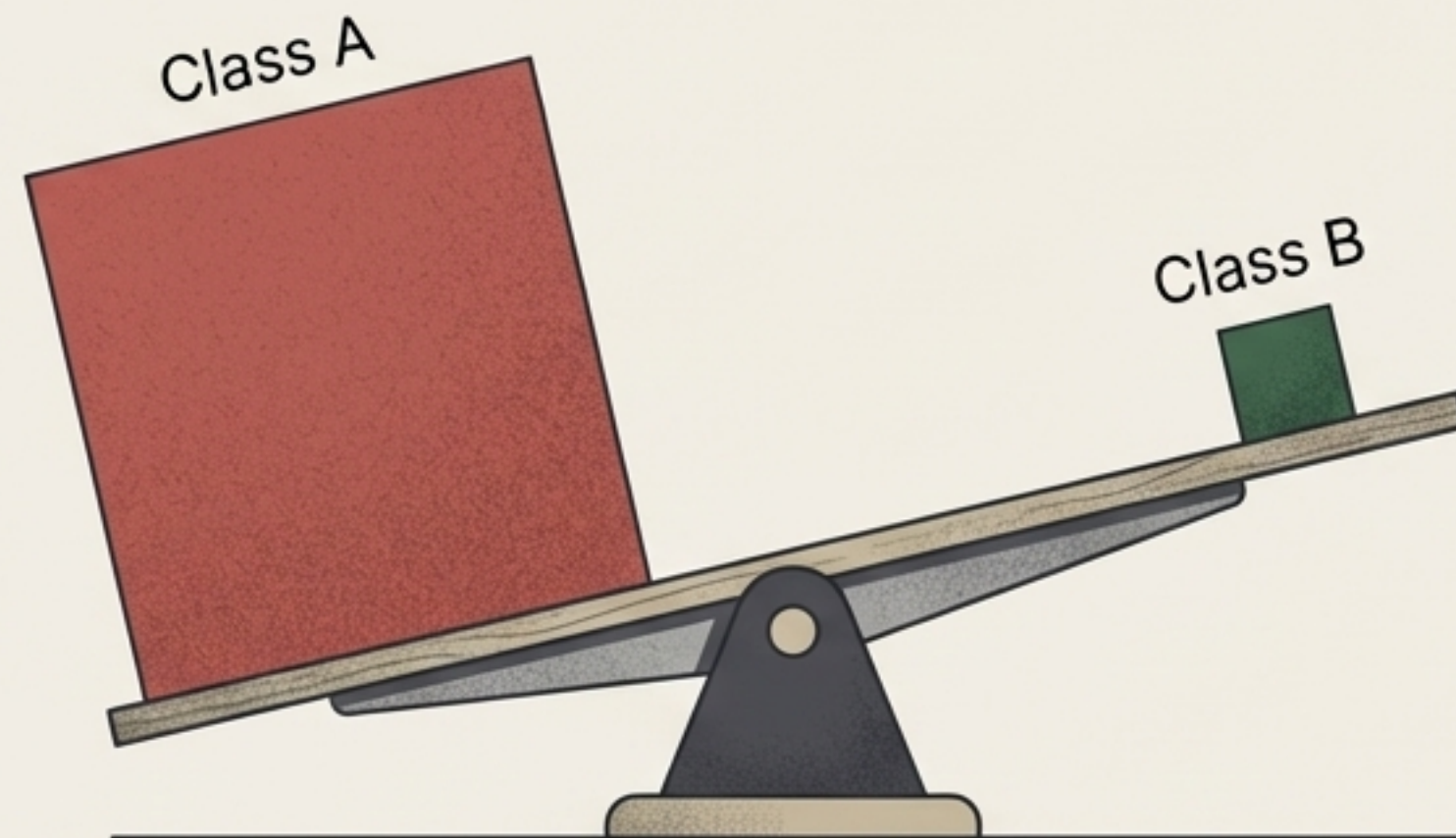
Blind Spots & Disadvantages

Without depth limits, trees will overfit to training noise. They also struggle to perform fairly when one class drastically outnumbered another.

1. Over-fitting



2. Imbalanced Data Sensitivity



The Ultimate Toolkit: CART

These exact geometric and mathematical principles power CART (Classification and Regression Trees). They require minimal data preparation, ignore the need for scaling or normalisation, and remain completely mathematically transparent.

